

Urban heat island in Southwark

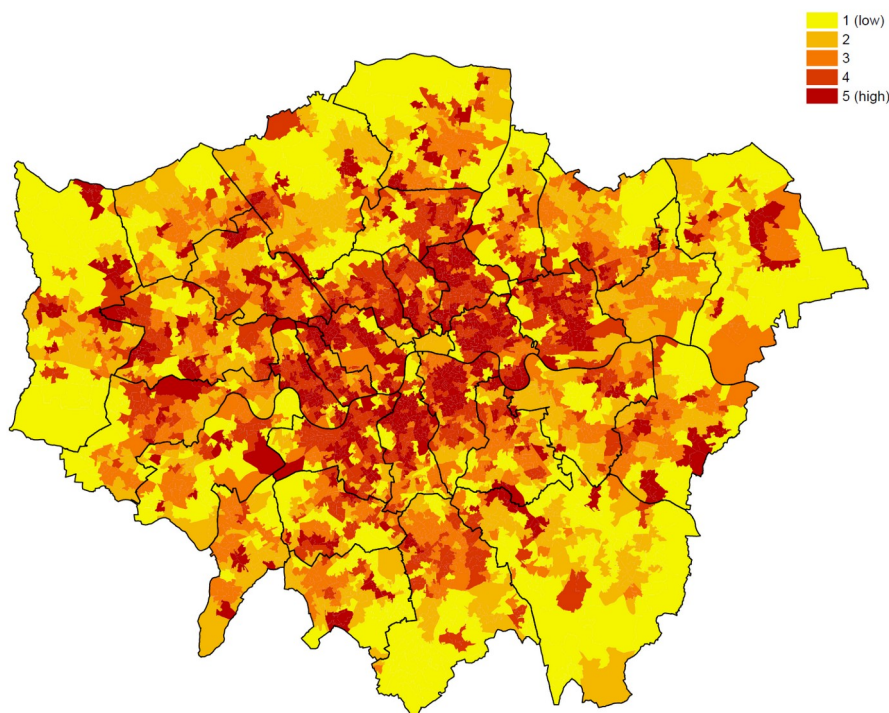
The urban heat island is the extra heat a city holds compared with the countryside around it. Roads, pavements and buildings absorb heat by day and trap it (London Climate Ready Partnership, n.d.). Southwark sits at the centre of London's heat island, with summer temperatures estimated around 4.5°C above rural areas (Southwark Council, 2025).

Evidence

- central London was up to 10°C warmer than rural areas in the 2003 heatwave, and the heat island can add 5 to 6°C to night-time temperatures (London Climate Ready Partnership, n.d.)
- in the July 2022 heatwave the heat island averaged 2.3°C and peaked at 7.2°C, strongest at night and in central London (Simpson, Brousse and Heaviside, 2024)
- it caused an estimated 141 deaths in Greater London over that heatwave, 38% of all heat-related deaths (Simpson, Brousse and Heaviside, 2024)
- 8% of Greater London (14 km²) is a persistent heat island, with mean summer surface temperature of 40.56°C or more over 2013 to 2023; clusters sit in east London, the western edge and south of the Thames (Krenz and Amann, 2025)
- Asian (39.10°C) and Black (39.00°C) residents face higher mean surface temperatures than White residents (38.57°C); children aged 4 and under face the highest of any age group (Krenz and Amann, 2025)
- the least well-off are exposed to surface temperatures 4.38°C higher than the most affluent (90th against 10th percentile) (Krenz and Amann, 2025)
- heat reduces walking accessibility in affected areas by 6 to 9%, hitting Asian and Black communities hardest (Krenz and Amann, 2025)
- the GLA's heat hazard mapping places Southwark among the central boroughs with the highest heat hazard, due to the heat island (Greater London Authority and Arup, 2024)
- green cover sets the local pattern: Kilburn and South Hampstead (38% vegetation) ran over 7°C hotter than nearby Regent's Park (89%) (Arup, 2023)

UHeatRank map

The GLA's heat hazard map, modelled by Arup with its UHeat tool, ranks every London LSOA from 1 (coolest) to 5 (hottest) by average summer air temperature, June to August 2018 to 2022. Each rank holds a fifth of LSOAs; averages range from 17.0°C to 19.5°C (Greater London Authority and Arup, 2024). Modelled air temperature is closer to what people feel than surface temperature. The ranks average day and night, so they understate heatwave peaks.



Contains National Statistics data © Crown copyright and database right 2023; Contains Ordnance Survey data © Crown copyright and database right 2023; European Commission, Joint Research Centre (JRC) 2023; Generated using Copernicus Climate Change Service Information 2023; Contains modified Copernicus Sentinel data 2023

Figure 1. Heat hazard by LSOA, London, June to August 2018 to 2022. Source: Greater London Authority and Arup (2024), Figure 16.

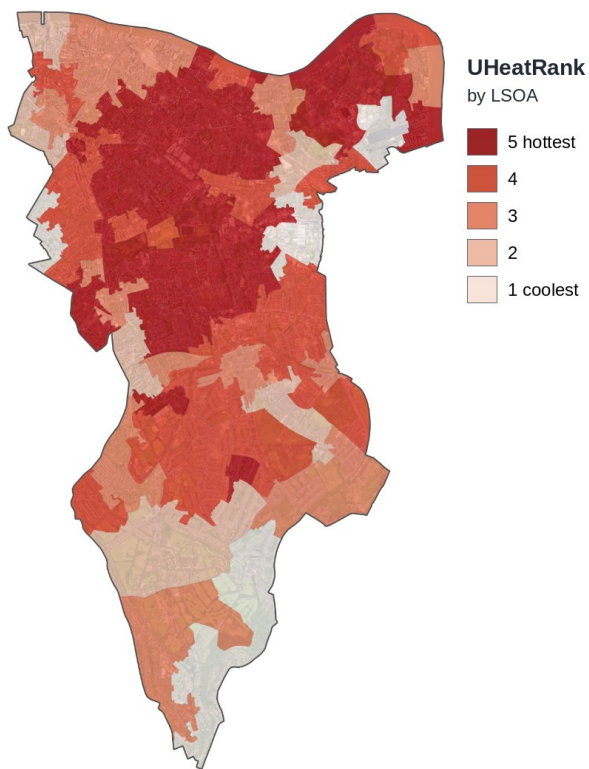


Figure 2. UHeatRank by LSOA, Southwark. Source: Greater London Authority and Arup (2024).

The hottest areas run through the north and centre of the borough. Walworth, Camberwell, Peckham and South Bermondsey combine the highest heat rank with the least green space and the highest deprivation (Southwark Council, 2025). The south of the borough and the larger parks rank lowest.

London heat island map

Krenz and Amann (2025) map persistent heat islands across London from Landsat summer imagery, 2013 to 2023, at 30 m resolution. Areas shown have mean surface temperature of 40.56°C or more.

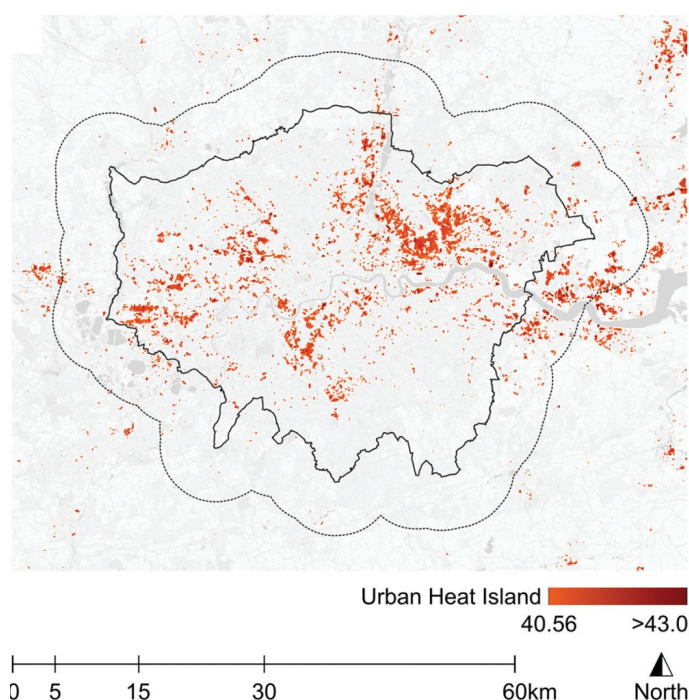


Figure 3. Urban heat island areas, Greater London, 2013 to 2023. Source: Krenz and Amann (2025), Figure 2b, CC BY 4.0.

Land surface temperature

The WRI Cool Cities Lab shows the same pattern at street scale. Hot-season surface temperatures across south London range from about 33°C to 43°C, with the Thames and large green spaces coolest (WRI, 2026). Surface temperature exceeds air temperature, so this shows pattern, not rank.

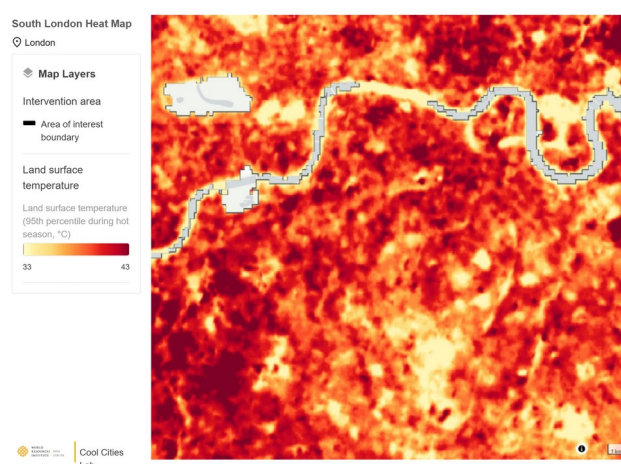


Figure 4. Land surface temperature, 95th percentile hot season (°C). Source: WRI (2026).

Works cited

Arup (2023) Urban Heat Snapshot. London: Arup. Available at: arup.com/insights/urban-heat-snapshot (Accessed: 29 September 2026).

Greater London Authority and Arup (2024) Properties Vulnerable to Heat Impacts in London: Prioritisation for Adaptation Interventions. London: Greater London Authority. Available at: london.gov.uk (Accessed: 29 September 2026).

Krenz, K. and Amann, L. (2025) 'Urban heat island effect: examining spatial patterns of socio-demographic inequalities in Greater London', *Cities & Health*, 9(5), pp. 901 to 924. Available at: doi.org/10.1080/23748834.2025.2489854.

London Climate Ready Partnership (n.d.) Heatwaves. Available at: climatelondon.org (Accessed: 29 September 2026).

Simpson, C.H., Brousse, O. and Heaviside, C. (2024) 'Estimated mortality attributable to the urban heat island during the record-breaking 2022 heatwave in London', *Environmental Research Letters*, 19(9), 094047. Available at: doi.org/10.1088/1748-9326/ad6c65.

Southwark Council (2025) Hot Weather and Health: Southwark's Joint Strategic Needs Assessment. London: London Borough of Southwark, Public Health Division.

World Resources Institute (WRI) (2026) Cool Cities Lab. Washington, DC: WRI Ross Center for Sustainable Cities. Available at: coolcities.wri.org (Accessed: 29 September 2026).